

Disease Diagnosis & Crop Care

AgriMitra AI Knowledge Base — Document 4 of 6 — Symptom-based Reference

How to Use This Guide

Match the observed symptom on the plant to the closest description below to narrow down the likely cause. This is a starting point for diagnosis, not a substitute for local plant pathology confirmation, especially before recommending any chemical treatment.

Paddy

Symptom	Likely Disease/Cause	General Care Direction
Spindle-shaped grey spots with brown borders on leaves	Blast disease	Avoid excess Nitrogen; ensure good field drainage; consider fungicide if severe
Yellowing and drying of leaves starting from the top, leaf tip	Bacterial leaf blight	Avoid water stagnation combined with high Nitrogen; use disease-resistant varieties
Brown, oval lesions with grey centers on leaves	Brown spot	Often linked to nutrient-deficient soils; balanced fertilization helps

Cotton

Symptom	Likely Disease/Cause	General Care Direction
Angular water-soaked spots on leaves	Bacterial blight	Use certified disease-free seed; avoid overhead irrigation
Sudden wilting of whole plant, vascular discoloration in stem	Fusarium wilt	Practice crop rotation; avoid continuous cotton cropping on the same land
Yellowing between veins, reddening of leaf edges	Nutrient deficiency (Nitrogen/Potassium)	Check soil for Nitrogen/Potassium deficiency via soil test

Chilli

Symptom	Likely Disease/Cause	General Care Direction
Upward curling, crinkling of leaves, stunted growth (spiral on top)	Viral infection (spiral top)	Remove and destroy severely infected plants; use virus-free seed
Sunken, dark circular spots on fruit with concentric rings	Anthracnose (fruit rot / die-back)	Avoid overhead irrigation; ensure good air circulation between plants
Sudden wilting of the whole plant, roots blackened	Damping off / root rot (soil-borne fungus)	Improve field drainage; avoid waterlogging around the root zone

Maize

Symptom	Likely Disease/Cause	General Care Direction
Long, cigar-shaped grey-green lesions on leaves	Tomato leaf blight	Practice crop rotation; avoid excess plant density
Small pustules (rust colored) on both leaf surfaces	Rust disease	Usually more severe in humid conditions; resistant hybrids recommended
Irregular, window-pane feeding damage on leaves	Fall armyworm (pest, not disease)	See Document 3 (Pesticide & Pest Management) for control

Groundnut

Symptom	Likely Disease/Cause	General Care Direction
Small, circular dark brown spots surrounded by yellow halo	Tikka leaf spot	Avoid dense sowing for better air circulation; remove crop debris
Yellowish-orange pustules mainly on the lower leaf surface	Rust	Often appears together with tikka leaf spot in humid weather
Sudden wilting with yellowing, plant easily pulled from soil	Early wilt (stem rot) (soil-borne fungus)	Avoid overwatering; use treated, disease-free seed

Sugarcane

Symptom	Likely Disease/Cause	General Care Direction
Red discoloration inside the stalk with a distinct alcohol-like smell	Red rot	Avoid ratooning from infected fields; use disease-free setts for planting
Yellowing and drying of leaves from margin inward, stunted growth	Phosphorus deficiency	Apply phosphorus fertilizer; test soil before assuming disease; check for nutrient deficiency with a soil/tissue test before assuming disease; check for nutrient deficiency
White, cottony fungal growth at the base of the stalk	Wilt disease complex	Improve drainage; avoid mechanical injury to the base during ratooning

General Crop Care Practices (All Crops)

- Inspect fields at least weekly during vegetative and reproductive stages to catch problems early
- Remove and destroy (do not compost) severely diseased plant material to reduce spread
- Avoid working in wet fields, since many fungal and bacterial pathogens spread easily on wet foliage
- Practice crop rotation with a non-host crop to break the disease cycle in the soil
- Maintain balanced fertilization — both excess Nitrogen and nutrient deficiency increase disease susceptibility
- Use certified, disease-free seed or planting material wherever available

Note for verification: Visual symptoms can overlap between diseases, nutrient deficiencies, and pest damage. Where possible, confirm diagnosis with a local Krishi Vigyan Kendra (KVK) or plant pathology lab before recommending a specific treatment, especially for high-value or large-area crops.